Marketing Analytics

Generative AI, Consumer Choice, and Regulation

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Session 14

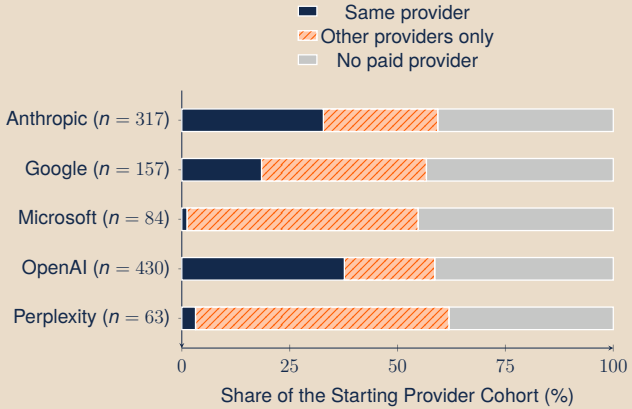
A Subscription-Growth Decision

- A firm wants to increase paid renewals.
- The evidence includes subscription histories, interfaces, and customer outcomes.
- A higher renewal rate can arise through improved value or greater cancellation friction.

Public Attention and Paid Demand

- Pageviews measure attention to selected pages.
- Paid subscriptions measure a specific payment relationship.
- User, provider, and category denominators differ.

A Fixed Starting Cohort

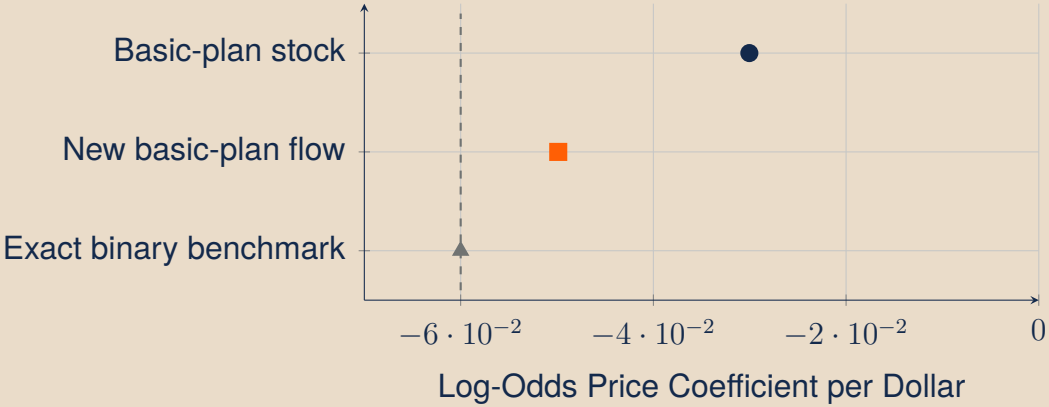


Entirely synthetic subscription panel. Provider labels identify simulated options.

Subscriber Stocks and New Flows

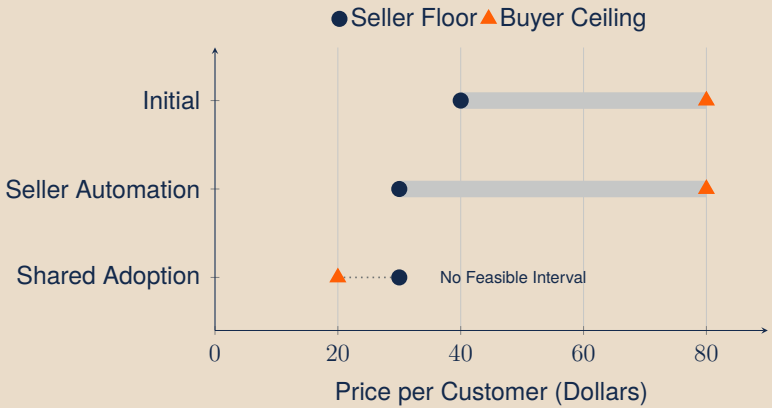
- A subscriber stock includes past acquisition and survival.
- A new-subscription flow can include first purchases and returns.
- A price coefficient depends on the outcome and behavioral model.

A Known Model Is a Useful Verification Case



Synthetic demand exercise with a known binary-model benchmark.

Seller Costs and Customer Alternatives

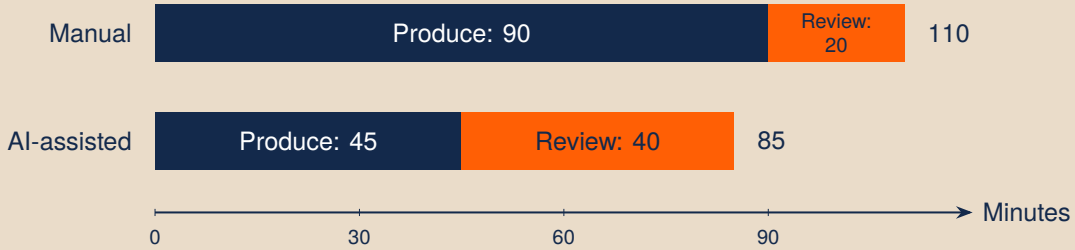


Hypothetical dollars per customer, conditional on 50 sales.

AI Workflow Quality

- Measure the completed analysis and its validity.
- Include generation, review, and repair time.
- Use comparable tasks and independent quality criteria.

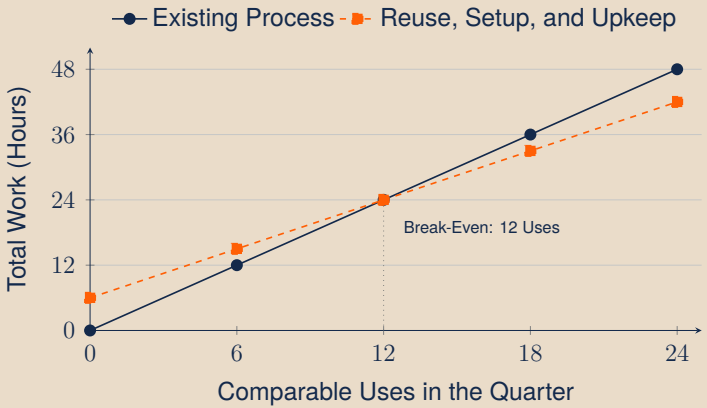
Review Time Is Part of Production



Drafting time falls 50%; total time falls 22.7%.
Task and completed-work quality held fixed.

Synthetic workflows meet the same stated quality requirement.

Reuse Has a Break-Even Workload



Hypothetical equal-quality work: six setup hours, with adaptation included.

Model Updates and Unresolved Tasks

- Identify the changed model, prompt, tool, data, or evaluation rule.
- Compare paired cases and fresh tasks using the same criteria.
- Include regressions, review costs, and customer consequences in the decision.

Residual Risk and Scale

- An expired research category can still allow residual failures.
- Expected failure loss depends on uses, failure probability, and consequences.
- Verify that references exist and support the claims made.

A Choice Procedure Can Change Renewal

- A direct offer and an offer preceded by questions can produce different choices.
- The customer still makes the final selection.
- An independent value measure determines whether an additional renewal benefits that customer.

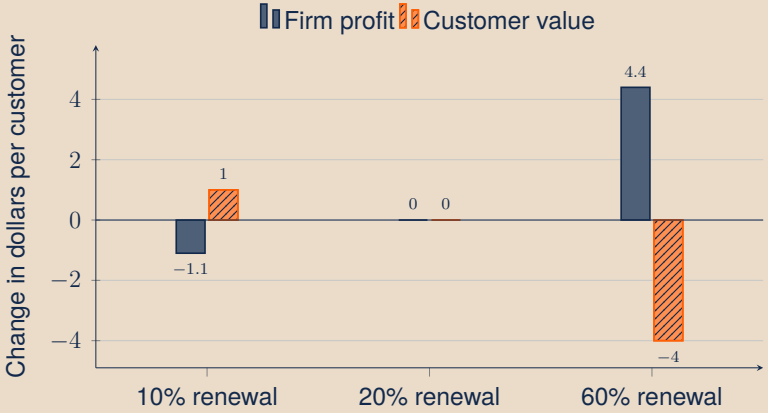
Opposed Interests

Suppose renewal yields firm margin $\nu > 0$ and costs the customer $\sigma > 0$ relative to declining.

$$\Delta\Pi = \nu(s - s_0), \qquad \Delta V = -\sigma(s - s_0)$$

A higher renewal probability raises profit and lowers customer value under these stated stakes.

Profit and Customer Value Can Move Oppositely



Synthetic probabilities and fixed stakes. Direct renewal is 20%.

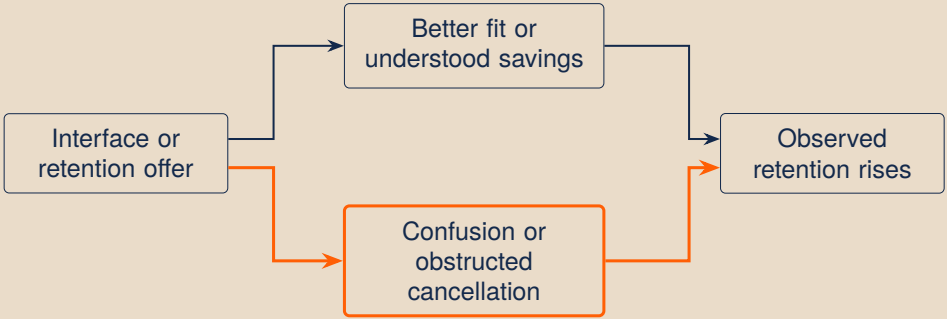
Experimentation Selects the Commercial Winner

- A fixed set of procedures has stable renewal probabilities.
- Independent randomized samples estimate each probability.
- With opposed interests, accurately selecting the largest rate also selects the largest loss in that set.

A Cancellation Evaluation

- Include a direct-cancellation comparison.
- Record whether optional retention content can be refused before exposure.
- Measure completion, elapsed time, subsequent charges, and customer value.

Retention Leaves the Mechanism Unresolved



Compare intent, understanding, effective exit, and later charges.
The same retention rate can have different customer consequences.

The same payment outcome can follow valuable service or obstructed exit.

Federal Cancellation Policy: Current Status

- July 2025: the Eighth Circuit vacated the 2024 FTC rule.
- February 2026: the FTC restored the earlier prenotification rule.
- March 2026: an advance notice sought input on possible amendments.

Cancellation Rules Differ by Jurisdiction

- California strengthened its Automatic Renewal Law on July 1, 2025.
- New York City’s separate rule took effect October 1, 2026.
- Identify covered transactions, enrollment media, and sector exceptions.

Research Frequency and Policy Importance

- Social influence leads the selected literature clusters.
- Exit friction can affect many customers even with few papers in the sample.
- Choose safeguards using mechanism, exposure, and customer consequences.

Pricing, Reviews, and Delivery

- Price-display rules require attention to the covered industry.
- Reviews presented as customer experiences require truthful provenance.
- Housing-ad delivery can create discrimination concerns beyond targeting choices.

Pricing Language and Regulatory Evidence

- Shelf infrastructure, price rules, and consumer effects are separate objects.
- The FTC’s January 2025 report describes research on pricing intermediaries.
- Specify the practice and applicable authority before making a legal claim.

A Welfare Comparison

Synthetic stakes distinguish profit from customer value.

```
direct, selected = 0.20, 0.60
firm_margin, customer_loss = 11, 10
firm_gain = firm_margin * (selected-direct)
value_change = -customer_loss * (selected-direct)
assert round(firm_gain, 2) == 4.40
assert round(value_change, 2) == -4.00
```

Higher combined surplus can coexist with a customer loss under these assumptions.

Coding Exercise: A Marketing Evaluation

- Calculate reuse break-even and the conditional price interval.
- Compute profit and customer value for alternative retention procedures.
- Propose an interface change and the evidence needed to evaluate it.

A Decision With Consumer Outcomes

- State the commercial target and customer outcome.
- Identify the procedure the customer can accept, refuse, or revise.
- Connect the proposed action to current rules and supporting evidence.